Ivan Evdokimov, Ph.D.

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Professional Summary

Software Engineer with 5+ years of production experience building backend systems, cloud infrastructure, and ML pipelines. Delivers across the full stack — PostgreSQL-backed APIs, CI/CD automation, AWS-based MLOps — with 95%+ model accuracy on sensitive datasets.

Experience

Senior Researcher, University of Essex — Colchester, UK Oct 2026 — present

Design and implement LLM-based solutions to extract, transform, and generate meaningful analytical features from unstructured qualitative data, integrating language models into research and data-processing pipelines while accounting for data confidentiality and privacy requirements.

- Led the team of junior researchers in the pivotal study of the homelessness predictions problem in Essex county.

Research Software Engineer, University of Essex — Colchester, UK Nov 2025 — Aug 2026

- Built GitHub Actions/Vercel CI/CD pipelines with automated unit and integration testing and code review, supporting an AI-assisted development workflow and enabling weekly releases across a multi-team Saas platform.
- Deployed Dockerised end-to-end MLOps pipeline for a Saas platform, reducing deployment time to 2 days and enabling continuous production monitoring.
- Developed a mobile app with OpenAI API integration for 20 field technicians, automating inspection reports and saving 3 weeks/year of manual data collection.
- Led UX design and user testing for the field-technician Java app, iterating on interface and workflow based on feedback from 20 field technicians to improve adoption and reduce reporting friction.

Data Engineer, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed classification PyTorch, scikit-learn, BERT models using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment in line with ONS statistical disclosure control (SDC) methodology, achieving 95%+ F1 accuracy on sensitive social science datasets, integrating and linking training data from multiple external sources.
- Deployed and orchestrated ML retraining pipelines using AWS Lambda, Step Functions and infrastructure-as-code practices, with monitoring for drift and accuracy to trigger automated retrains — reducing manual intervention by 50%.
- Built ETL pipelines to extract, transform, and query PostgreSQL-hosted socioeconomic data of 100K+ records as model inputs for automated metadata classification, improving throughput and classification accuracy at scale.
- Employed density-based clustering (DBSCAN) on questionnaire datasets to segment studies according to concentration of disclosive household attributes.
- Developed a LangGraph-based agent-to-agent (A2A) workflow with hallucination checks to automate exploratory data analysis (EDA) — including dataset loading, statistical description, and property derivation — reducing the time independent researchers spent on manual data analysis.

Research Officer & Teaching Assistant, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime 8x for collaborations with other universities in the UK.
- Delivered lectures and labs in Data Structures, C/C++, and Machine Learning to 100+ undergrad and postgrad students, achieving an average student feedback score of 4.8/5.

Projects

Chat-LLM github.com/ivan020/ChatLLM

- Built a self-hosted RAG pipeline over a Raspberry Pi-hosted Ollama LLM using LangChain, pgvector, and all-MiniLM-L6-v2 embeddings, indexing past stream transcripts to allow new viewers query past content in real time via Twitch chat.

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Built a Dockerised real-time portfolio tracker in Golang with buy/sell mechanics, a PostgreSQL (Supabase) backend recording all transactions, and a live leaderboard — using Twitch chat as the user interface.
- Developed a TypeScript React frontend with real-time portfolio visualisation, interactive performance graphs, and transaction history, serving 120+ monthly users.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Built a Dockerised Golang REST API that uses OCR to extract text from Magic: The Gathering card photos, identifies the card via LLM, and returns real-time price data from a PostgreSQL-backed catalogue, displayed via the TypeScript React frontend.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Developed a transfer learning framework for Bayesian model averaging in financial forecasting, first-authored a peer-reviewed CS/ML journal publication, and co-authored international conference papers; released an open-source package with external

adoption.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Sole-authored a modular Python agent-based simulation modelling household consumption, employment, and borrowing behaviour under dynamic and negative interest rates, with an extensible architecture supporting additional firm and household modules.